

冒頓: the three readings against the three candidates

Working note, 3 September 2026. Not part of the paper or the supplement.

The first character is not in dispute

冒 has two Later Han readings, mək and mou^c. Both Schuessler and Pulleyblank assign the first to this compound, and Pulleyblank names it: *covet; Modun, founder of the Xiongnu empire*, K1062a, EMC mak. Every candidate below therefore starts from mək, whose written -k is what allows one character to carry two source syllables.

The second character: three assignments, one row each

READING OF 頓	WHAT IT IS, AND WHO ASSIGNS IT HERE	BÖGÜ TIN BÖGÜ + TIN "SPIRIT, BREATH"	BÖGÜ TUĞ BÖGÜ + TUĞ "HORSEHAIR STANDARD"	BATATUR ONE WORD, "HERO"
tuən ^c nasal coda	Row 1 of Schuessler's table, the character's primary reading. Assigned here by the author, and printed in Appendix A. Schuessler's own note for this name points to row 2 instead, so the choice of the primary reading is a position that has to be argued rather than assumed.	fits a source dental nasal written by a dental nasal is the commonest cell in the corpus, 63 of 159 (step 34). Nothing further needed.	fails a source velar stop written by a dental nasal: 0 of 20 (step 34). Manner is the one thing that does not switch.	fails source - r against a written - n: 1 of 37 (step 23)
tuət dental stop	Row 2 of Schuessler's table. Assigned to this name by Schuessler, pp. 209 and 361, in both places as "Alternate reading in the name Mao-dun, then read Mo-du LH mək-tuət".	fails a source nasal written by a stop: 0 of 159 (step 34). A hard zero, not a low rate.	permitted, minority a source velar stop written by a dental stop: 2 of 20, 10% (step 34). Place switches in 17 of 58 written codas, so this is a real if uncommon spelling. The loanword objection still stands.	fits source - r written - t: 14 of 37, 38% (step 23)
douk velar stop	Not a reading of 頓. The 毒 syllable substituted by the Chinese commentary, 頓音毒. Reported by Pulleyblank 1991 at K1062a as EMC dawk. A tradition about how to read the character, which is a different kind of claim from a reconstruction of it.	fails a source nasal written by a stop: 0 of 159 (step 34)	fits phonologically both codas written and same place, the -k carrying the -ğ: 7 of 20, 35% (step 34), nothing unwritten. But circular: Doerfer 969 derives Turkic <i>tuğ</i> from Chinese 纛 douk itself, and Franke dates the Chinese word to the <i>Zhouli</i> , no later than the 2nd century BC.	fails source - r would need a velar coda: 0 of 37 (step 23)

Each reading admits exactly one or two candidates, and they are not the same ones. The reading printed in Appendix A is the only one of the three on which the author's own proposal survives. That does not make the assignment wrong: taking a character's primary reading is a defensible default, and it is the author's to make. But the coincidence is exactly what §6.3 measures, so the reason for preferring row 1 over the row Schuessler names for this name has to be stated in the paper rather than left implicit. On row 2, the one Schuessler assigns, *Bayatur* is better supported than *Bögü Tuğ* by roughly four to one and carries no historical objection.

What step 34 measured, and it is a rule worth having on its own

The question was whether a written coda has to match the place of articulation of the source coda. It does not, but it does have to match the manner.

SOURCE CODA	OPEN	-K	-T	-P	-ŋ	-N	-M
velar stop	11	7	2	0	0	0	0
dental stop	16	9	31	0	0	1	0
labial stop	5	1	5	2	0	0	0
any nasal	9	0 as a stop			150 as a nasal		

Place is negotiable, manner is not. Of 58 stop-final source syllables whose coda was written, 41 kept the place and 17 changed it, 29%. But a source nasal was written as a stop **0 times in 159**, and a source stop as a nasal once in 58. The 36% of stop-final syllables written with an open character reproduces Table 7's 30% on a different subset.

That asymmetry is what decides the matrix above. Every cell marked "fails" for *Bögü Tın* fails on manner, which never bends. The cell that changed for *Bögü Tuğ* changed because place does.

Costs every candidate pays equally, so they decide nothing between them

- **One character for two source syllables**, with a written stop coda doing the compressing: 47 of the 1,017 verified pairs, **4.6%**. *Bögü Tın*, *Bögü Tuğ* and *Bayatur* all require it, for *bögü* and *baya* respectively.
- **A Chinese m- character for a source b-**: 3 of 22 in the Later Han layer, **13.6%** (§8.1), corrected from the period-mixed 0.4%.
- **The vowels are free**. §7 puts vowel recovery at 50.0% with the interval straddling one half, so no candidate is preferred or refused on a vowel.

What is not measured yet

- Only three Turkic words fit the $\text{tu}\text{ə}\text{t}$ slot at all: *tat* "foreigner", *tüt* "larch", *tut* "rust". So on Schuessler's reading there is no Turkic compound and *Bayatur* stands alone. That search was ad hoc and needs a numbered run.
- Step 23's 37 liquid opportunities remain the weakest sample in the table: every *Bayatur* cell rests on them.

Sources: Schuessler 2009, *Minimal Old Chinese and Later Han Chinese*, pp. 209 and 361 · Pulleyblank 1991, *Lexicon of Reconstructed Pronunciation*, K1062a · Doerfer, *TMEN* II 969 and IV 969, with Menges 1968: 169 and Franke's letter of 23 March 1966 · Clauson 1972, s.v. *tu:ğ* · steps 20, 22, 23, 24 and 31 of this project.